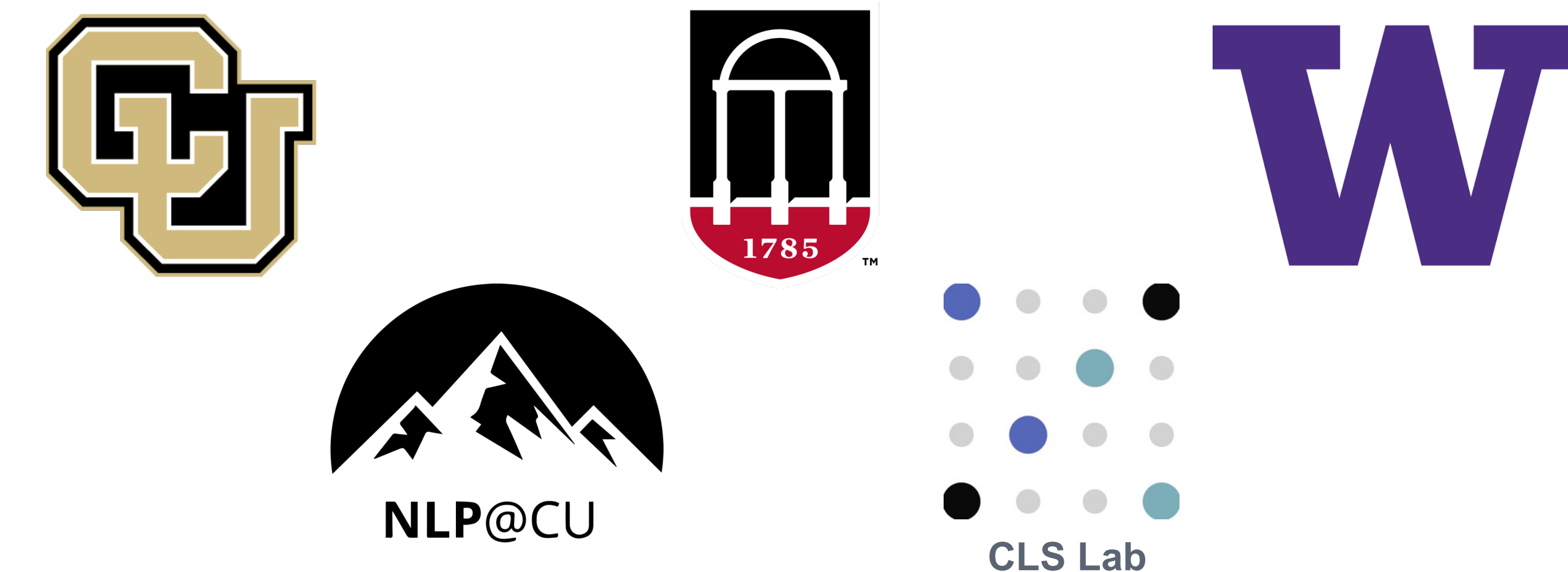


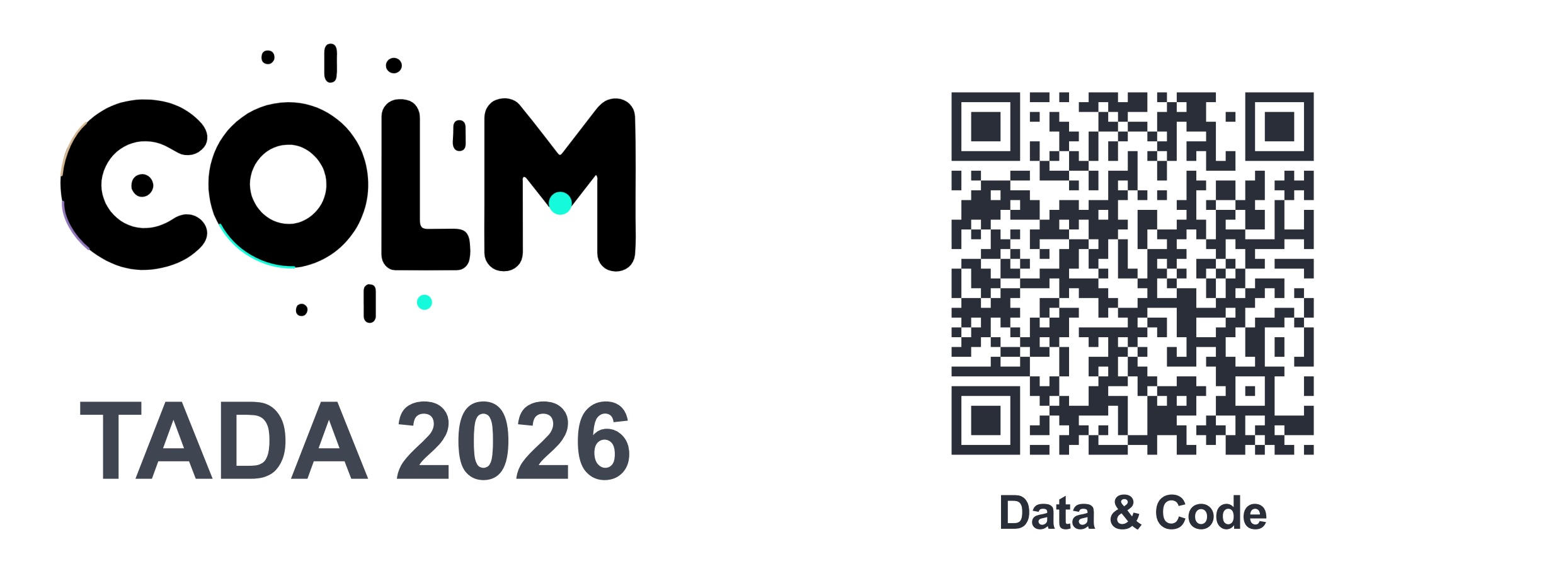


The Garden of Forking Prompts

How Users Explore Narrative Space in Story Generation

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Our contributions

Inspired by Jorge Luis Borges's 1941 story "The Garden of Forking Paths," we study the paths users take when generating fiction with chatbots. From ~4.8M WildChat conversations, we identify ~275K story prompts and reconstruct users' exploration trees using lexical and temporal signals. From these trees, we formalize a taxonomy of actions users take while navigating narrative space.

WildStories

275,635 prompts, annotated for format, prompt components, and explicitness.

WildEdits

24,291 trees, 100,200 prompt pairs, 165,846 edit actions; WildGuard labels response refusals.

Edit Taxonomy

Open-coded from 106 real revisions: 4 directions × 14 targets for analysis and realistic prompt permutations.

DATA PROCESSING PIPELINE

Step 1: Extract & classify prompts

Extract prompts → Length + language → Classify using Qwen3.5

WildChat: 4.74M conversations → 2.56M eligible prompts
Human: n=100, P=.78, R=1.00, IAA α=.78

WildStories 275,635 story prompts

Step 2: Cluster near-duplicate prompts

word 3-grams → Jaccard ≥ 0.5 → Union-Find

Drop singletons. Human: n=100 balanced pairs; P=R=1.00; IAA=100%

22,999 clusters
cluster: coffee scene
cluster: dragon tale

Step 3: Infer users → merged_owner_id

Exact-prompt merge → Cluster co-occurrence

Shared prompt and cluster histories resolve hashed IPs to canonical users.

IP A, IP B → User
2 IPs → 1 user

Step 4: Build prompt trees

Sort by timestamp → Parent = most-similar earlier prompt

24,291 trees, similarity < 0.5 pruned

Step 5: Open-code an edit taxonomy + annotate pairs

Compact marked diff → Gemma4-31B → 4 directions × 14 targets

CHANGE

two girls **vietnamese men**
character culture

ADD

...soda machine , **it broke**
plot

EXTEND

...the end. **The prince woke...**
plot

REMOVE

raised in a **big-city**
backstory

WildEdits 100,200 edit pairs

Gemma 4 31B IT labels changed spans. Human: n=100, 91% agreement, IAA α=.75

Users frequently combine multiple different edit targets and directions

Write an extremely long story about a man named Steven. He's in a magical college that saves people who were about to die before their time by transporting them to the college where the person is transformed into a mythical being. Have the story start with Steven waking up a dorm after passing out in a house fire. He's confused on how he's alive and why he's an anthropomorphic dragon. He suddenly **(-R1) plot coughs** **(+A1) plot accidentally summons** fire **(+A1) plot at his fingertips** and **(-R1) plot freaks out** **(+A1) plot** suffers a PTSD induced flashback due to having PTSD around fire. **(+A2) plot** A nerdy elf woman named Ev who is his new roommate finds him and calms him down. Have the rest of the story be about him **(-R2) plot meeting a nerdy elf woman named Ev** and falling in love with **(-R3) wording her**. **(+A3) wording Ev**. Have 5 mind bogglingly long chapters. Do not introduce the characters or settings. Have Steven talk like a standard young adult. **(+A4) character description** Have Ev have boyish hobbies and be very chill. Have plenty of dialogue.

EDIT ACTIONS (4)

plot -R1 +A1

plot -R2 +A2

wording -R3 +A3

character description +A4

Edit trees reveal different exploration strategies: shallow exploration or deep refinement

(a) Make a Part 1 with a talk script scenario about Milo, Juniper, and SparrowBot on a mission but then something happens, something comes to attack them but then Milo goes into his shadow form mode.

(b) Write a short story about a girl who played in the gray ash of a distant eruption as it was falling

(c) Three of 24,291 prompt trees (paraphrased). Each branch links a prompt to its most-similar earlier version; links below 0.5 similarity were pruned.

Plot edits dominate, Extend is largely plot-centric

Target	ADD	REMOVE	CHANGE	EXTEND
Narrative				
Plot	18,415	5,345	24,498	18,670
Dialogue	1,873	610	3,085	109
Setting	2,738	615	2,890	78
Characters				
Description	11,660	2,890	8,881	188
Name	879	132	2,159	24
Gender	146	20	216	0
Culture	369	141	1,004	3
Substitution	45	33	2,714	0
Backstory	2,978	534	1,359	54
Fandom	307	98	1,389	2
Style				
Wording	5,181	3,504	15,537	1
Genre/Style	2,170	911	2,739	2
Meta				
Instructions	8,883	2,684	6,050	15
Structure	59	113	613	0

Edit targets are format specific

Target	Prose	Roleplay	Script	Narration
Narrative				
Plot	-0.06	0.11	0.09	-0.51
Dialogue	-0.73	0.90	0.67	-1.83
Setting	0.11	-0.18	-0.21	0.65
Characters				
Description	0.14	-0.30	-0.23	-0.84
Name	-0.07	0.64	-0.11	-0.17
Gender	0.10	-0.13	-0.18	—
Culture	-1.28	-1.75	1.30	-0.14
Substitution	0.02	-0.14	-0.01	-0.54
Backstory	0.10	-0.07	-0.23	0.36
Fandom	0.19	-0.52	-0.33	-1.00
Style				
Wording	0.24	-0.61	-0.46	0.01
Genre/style	0.06	-0.92	0.04	1.70
Meta				
Instructions	0.15	0.56	-0.69	1.39
Structure	0.00	0.86	-0.42	0.78

Edits by target and direction.

PMI between edit target and story format.

Key findings

- Story exploration routinely crosses conversations. 94.2% of linked prompts have a parent in another WildChat conversation. Prior works studying user-chatbot interactions only focus on turns in a single thread and miss this behavior.
- Explicit prompts sustain deeper exploration. 43.5% of story prompts are explicit (as opposed to 8.5% of overall prompts) and trees that contain explicit prompts are on average 1.78× larger; and they remain so even after accounting for refusals.
- User activity is highly skewed and tree distribution is long tailed. Users: median 1 tree (IQR 1–2; max 5,560). Trees: median 3 prompts (IQR 2–7; max 1,811).
- Editing only yields modest personalization. On average, deepest leaves gain only +0.039 median story specificity; prompt specificity does not change.

Takeaways

- Observed edits can guide the design for realistic benchmark variants. Taxonomy-guided permutations best match real revisions in our case study (L2 0.15 versus 0.28 and 0.74), validating our taxonomy and enabling further work in story generation benchmarks.
- Observations of real user-chatbot interactions are crucial. With chatbot usage skyrocketing, analysis of these user interactions is paramount. Findings from these studies can help inform the design of better, responsible technology.
- Working with real user-chatbot interactions is difficult. Large scale user-chatbot interaction data is messy. Datasets are often skewed by prolific users and extraordinary care is required when sampling, deduping, filtering, categorizing and clustering such data.

With support from: